

Lab Report No 10
Understanding and practice of Step Motor
Control and Photo Interrupter Control.
Embedded Systems



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Lab Task No 2:

Solution:

Brief description (3-5 lines)

Write an assembly language code for the program in exercise#1 using the RR instruction.

The code

<pre>ORG 0000H ; Starting address of the program MAIN: MOV A, #0FEH ; Initialize accumulator with 1111 1110 (binary) MOV P1, A ; Send initial value to Port 1 (LEDs) LOOP: ACALL DELAY ; Call delay subroutine ACALL DELAY_5US ; Call 5-microsecond delay RR A ; Rotate accumulator right MOV P1, A ; Update Port 1 with the rotated value SJMP LOOP ; Repeat the loop ; Delay subroutine to create a visible effect of rotation DELAY: MOV R0, #200 ; Outer loop count D1: MOV R1, #255 ; Inner loop count D2: DJNZ R1, D2 ; Decrement R1 until it reaches 0 DJNZ R0, D1 ; Decrement R0 until it reaches 0 RET ; Return from subroutine ; Subroutine for 5 microsecond delay (approximation based on 8051 clock cycle) DELAY_5US: NOP ; 1 machine cycle (1 microsecond for 12 MHz clock) NOP ; 1 machine cycle NOP ; 1 machine cycle NOP ; 1 machine cycle NOP ; 1 machine cycle RET ; Return from subroutine END</pre>

The results (Screenshot)

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creating hex file from ".\Objects\stepmotor"...\n".\Objects\stepmotor" - 0 Error(s), 3 Warning(s).\nBuild Time Elapsed: 00:00:00
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Conclusion

In conclusion, the precise angular movement is possible with the step motors in response to externally supplied electrical pulse sequences that are turned on at predetermined intervals. The presence or position of an object is detected by a photo interrupter when the light beam is interrupted between the emitter and receiver. The combination of step motor control and photo interrupter sensing leads to excellent systems for precisely moving mechanical structures and for verifying their position in real time. All these technologies play a critical role in robotics, automation, and precision engineering applications and lend reliability as well as accuracy to motion and detection capabilities.